CONTEMPORARY HEALTH ISSUES

Bulky First-Year Study Notes – Semester One

Prepared as an exam-oriented study guide for a first-year student at Karatina University

Study note: These notes are educational. For clinical decisions, use current guidance from qualified health professionals and relevant Kenyan health authorities.

TABLE OF CONTENTS

1. Introduction to Contemporary Health Issues

2. Dimensions and Concepts of Health

3. Determinants of Health

4. Communicable Diseases

5. HIV/AIDS and Sexually Transmitted Infections

6. Malaria and Other Vector-Borne Diseases

7. Tuberculosis

8. Emerging and Re-emerging Infections

9. Non-Communicable Diseases

10. Hypertension and Cardiovascular Disease

11. Diabetes Mellitus

12. Cancer

13. Mental Health

14. Substance Use and Abuse

15. Sexual and Reproductive Health

16. Maternal, Newborn and Child Health

17. Nutrition and Nutritional Disorders

18. Environmental Health

19. Water, Sanitation and Hygiene

20. Occupational Health and Safety

21. Lifestyle and Health

22. Poverty, Inequality, Gender and Health

23. Disability and Ageing

24. Health Systems, Primary Health Care and Health Promotion in Kenya

25. Health Education and Disease Prevention

26. Contemporary Health Challenges and Future Directions

Revision Questions and Model Answers

1. Introduction to Contemporary Health Issues

Contemporary health issues are health problems that affect individuals and communities in the present period and are influenced by biological, social, economic, environmental, cultural and technological changes. They include communicable diseases, non-communicable diseases, mental-health problems, malnutrition, substance use, environmental hazards, reproductive-health concerns and emerging infections.

• Health: a state of physical, mental and social well-being, rather than merely the absence of disease.

• Illness: the person's experience or perception of not being well.

• Disease: a pathological condition that interferes with normal body functioning.

• Public health: organized efforts to prevent disease, prolong life and promote health through population-level action.

• Health promotion: enabling people to increase control over and improve their health.

2. Dimensions and Concepts of Health

Health is multidimensional. The physical dimension concerns body functioning and freedom from preventable disease. Mental health involves emotional balance, coping and effective functioning. Social health concerns relationships and participation in society. Other useful dimensions include emotional, spiritual, environmental and occupational well-being.

• Health is dynamic and changes over time.

• Health exists on a continuum from severe illness to optimal well-being.

• Individual behaviour matters, but health is also shaped by wider social and environmental conditions.

• Holistic healthcare considers the whole person rather than one symptom or disease.

3. Determinants of Health

Determinants are factors that influence health outcomes. Major determinants include income and employment, education, housing, food security, social support, physical environment, access to healthcare, genetics, age and sex, culture, personal behaviour and government policy.

• Socioeconomic status can influence nutrition, housing and access to treatment.

• Education can improve health literacy and informed decision-making.

• Unsafe water and sanitation increase infectious-disease risk.

• Policies affecting tobacco, alcohol, food, transport, housing and healthcare can influence population health.

4. Communicable Diseases

Communicable diseases are illnesses caused by infectious agents that can spread directly or indirectly between people, animals or the environment. Agents include bacteria, viruses, fungi, protozoa and parasites.

• Modes of transmission: direct contact, droplets, airborne spread, contaminated food or water, blood and body fluids, vectors and contaminated objects.

• Prevention: vaccination, hand hygiene, safe water, sanitation, food safety, ventilation, vector control, early diagnosis and appropriate treatment.

• Chain of infection: infectious agent → reservoir → portal of exit → mode of transmission → portal of entry → susceptible host.

• Breaking any link in this chain can reduce transmission.

5. HIV/AIDS and Sexually Transmitted Infections

HIV attacks the immune system. Without effective treatment, advanced HIV infection can lead to AIDS. HIV is transmitted mainly through infected blood, semen, vaginal fluids and breast milk under circumstances that allow transmission. It is not spread through ordinary social contact such as hugging or sharing utensils.

• Prevention includes safer sexual practices, HIV testing, prevention of mother-to-child transmission, sterile injection equipment and appropriate biomedical prevention.

• STIs include infections such as syphilis, gonorrhoea, chlamydia, trichomoniasis and genital herpes.

• Untreated STIs may cause infertility, pregnancy complications and increased HIV transmission risk.

• Stigma can discourage testing and treatment; confidentiality, respectful services and accurate information are important.

6. Malaria and Other Vector-Borne Diseases

Malaria is caused by Plasmodium parasites and is transmitted by infected female Anopheles mosquitoes. Common features include fever, chills, headache and weakness, although symptoms can vary and severe disease can develop rapidly.

• Prevention: insecticide-treated nets, indoor residual spraying where recommended, reducing mosquito breeding sites and prompt diagnosis and treatment.

• Vector control reduces contact between humans and disease-carrying organisms.

• Other vector-borne diseases include dengue, yellow fever and some forms of viral encephalitis.

7. Tuberculosis

Tuberculosis (TB) is an infectious disease caused mainly by Mycobacterium tuberculosis. Pulmonary TB commonly spreads through the air when an infectious person coughs, speaks or sneezes.

• Possible symptoms include prolonged cough, fever, night sweats, weight loss and fatigue.

• Diagnosis and treatment should follow national clinical guidance.

• Adherence to the prescribed treatment regimen is important to prevent treatment failure and drug resistance.

• Good ventilation, respiratory hygiene, early diagnosis and contact investigation can help reduce spread.

8. Emerging and Re-emerging Infections

Emerging infections are newly recognized or rapidly increasing infections; re-emerging infections are diseases that were previously controlled but are increasing again. Drivers include urbanization, travel, climate change, ecological disruption, antimicrobial resistance and changes in human-animal contact.

• Preparedness includes surveillance, laboratory capacity, infection prevention, risk communication and rapid response.

• One Health recognizes connections among human, animal and environmental health.

9. Non-Communicable Diseases

Non-communicable diseases (NCDs) are generally long-lasting conditions that are not transmitted from person to person. Major groups include cardiovascular diseases, cancers, chronic respiratory diseases and diabetes.

• Risk factors include tobacco use, harmful alcohol use, unhealthy diet, physical inactivity and air pollution.

• Prevention combines individual action with supportive policies and healthy environments.

• Early detection and long-term management can reduce complications and premature mortality.

10. Hypertension and Cardiovascular Disease

Hypertension is persistently elevated blood pressure. It is often asymptomatic, which is why routine measurement is important. Cardiovascular diseases include heart disease and stroke.

• Risk reduction: reduce salt intake, maintain a healthy weight, be physically active, avoid tobacco, limit harmful alcohol use and follow prescribed treatment.

• Uncontrolled hypertension can damage the heart, brain, kidneys and blood vessels.

• Stroke warning signs can include sudden facial weakness, arm weakness and speech difficulty; urgent medical attention is required.

11. Diabetes Mellitus

Diabetes is a chronic metabolic disorder characterized by elevated blood glucose. Type 1 diabetes results from autoimmune destruction of pancreatic beta cells; type 2 diabetes involves insulin resistance and progressive impairment of insulin secretion; gestational diabetes occurs during pregnancy.

• Symptoms may include excessive thirst, frequent urination, unexplained weight loss, fatigue and blurred vision.

• Management may involve healthy eating, physical activity, medicines and regular monitoring.

• Long-term high blood glucose can damage eyes, kidneys, nerves, heart and blood vessels.

12. Cancer

Cancer is a broad group of diseases in which abnormal cells grow uncontrollably and may invade or spread to other tissues. Cancer risk is influenced by genetics, age, infections, tobacco, alcohol, diet, obesity, radiation and environmental exposures.

• Prevention includes tobacco control, vaccination against cancer-causing infections where appropriate, healthy lifestyle choices and reduction of harmful exposures.

• Screening is useful for selected cancers when recommended for the relevant population.

• Early diagnosis improves the possibility of effective treatment for many cancers.

13. Mental Health

Mental health is a state of well-being that supports coping with normal stresses, learning, working and contributing to the community. Mental-health conditions may include depression, anxiety disorders, psychotic disorders and substance-use disorders.

• Warning signs may include persistent sadness, severe anxiety, major changes in behaviour, social withdrawal, sleep or appetite changes and impaired functioning.

• Protective factors include supportive relationships, adequate sleep, physical activity, meaningful activity and access to appropriate care.

• Stigma and discrimination can delay help-seeking.

• Suicidal thoughts or immediate danger require urgent support from emergency services or a qualified health professional.

14. Substance Use and Abuse

Psychoactive substances can affect mood, cognition and behaviour. Harmful use may produce dependence, injuries, family difficulties, poor academic performance and chronic health problems.

• Commonly discussed substances include alcohol, tobacco/nicotine, cannabis and other drugs.

• Prevention includes accurate education, supportive families and communities, healthy coping strategies and accessible treatment.

• Dependence is a health condition and should be approached without stigma.

15. Sexual and Reproductive Health

Sexual and reproductive health includes physical, emotional, mental and social well-being related to sexuality and reproductive processes. It involves informed choices, respectful relationships, prevention of STIs, contraception, fertility care, pregnancy care and protection from violence and coercion.

• Comprehensive health education supports informed decision-making.

• Contraceptive methods vary in effectiveness, benefits and possible side effects; method choice should be informed and voluntary.

• Pregnancy and childbirth require access to skilled healthcare and timely referral when complications occur.

16. Maternal, Newborn and Child Health

Maternal and child health focuses on the health of women during pregnancy, childbirth and the postpartum period and on the survival and healthy development of newborns and children.

• Antenatal care can identify and manage risks.

• Skilled birth attendance and emergency obstetric care reduce preventable complications.

• Breastfeeding, immunization, appropriate complementary feeding, growth monitoring and infection prevention support child health.

• Children need protection from abuse, neglect and preventable injuries.

17. Nutrition and Nutritional Disorders

Nutrition is the process by which the body obtains and uses nutrients for energy, growth, repair and normal functioning. Malnutrition includes undernutrition, micronutrient deficiencies, overweight and obesity.

• Macronutrients: carbohydrates, proteins and fats.

• Micronutrients: vitamins and minerals.

• Balanced diets should contain adequate variety and appropriate quantities of nutrient-rich foods.

• Nutrition problems may result from poverty, disease, food insecurity, misinformation, poor dietary patterns or increased nutritional needs.

18. Environmental Health

Environmental health examines how physical, chemical and biological factors in the environment affect human health. Important issues include air pollution, unsafe water, poor sanitation, waste, climate change, chemical exposure and unsafe housing.

• Safe drinking water and sanitation reduce many infectious diseases.

• Indoor and outdoor air pollution can contribute to respiratory and cardiovascular illness.

• Proper waste management reduces injuries, pests, contamination and environmental pollution.

• Climate change can affect heat exposure, food systems, water availability and vector-borne disease patterns.

19. Water, Sanitation and Hygiene

WASH refers to water, sanitation and hygiene. Handwashing with soap at critical times, safe drinking water, hygienic sanitation and safe food handling are major public-health measures.

• Critical handwashing moments include after using the toilet and before preparing or eating food.

• Water should be protected from contamination from source through transport and storage.

• Communities benefit from sanitation systems that safely contain and manage human waste.

20. Occupational Health and Safety

Occupational health aims to protect workers from hazards arising from their work. Hazards can be physical, chemical, biological, ergonomic or psychosocial.

• Risk assessment identifies hazards, who may be exposed, the likelihood and severity of harm, and appropriate controls.

• Controls should prioritize elimination or reduction of hazards, safe systems of work, training and appropriate personal protective equipment.

• Workplace health also includes mental well-being, fatigue management and prevention of violence and harassment.

21. Lifestyle and Health

Lifestyle factors can strongly affect long-term health. Physical inactivity, unhealthy diet, tobacco use, harmful alcohol use, inadequate sleep and chronic stress are important modifiable risks.

• Regular physical activity supports cardiovascular, metabolic and mental health.

• Healthy eating emphasizes dietary diversity and moderation of highly processed foods high in salt, free sugars or unhealthy fats.

• Sleep supports physical recovery, learning and emotional regulation.

• Behaviour change is easier when healthy choices are affordable, accessible and socially supported.

22. Poverty, Inequality, Gender and Health

Health is closely related to social and economic conditions. Poverty can limit access to nutritious food, safe housing, education and healthcare. Inequality can create unequal exposure to risks and unequal access to opportunities.

• Gender norms can influence exposure to violence, healthcare access, reproductive choices and health-seeking behaviour.

• Health equity means reducing avoidable and unfair differences in health outcomes.

• Community participation helps ensure health interventions reflect local needs and realities.

23. Disability and Ageing

Disability can result from interaction between health conditions and environmental or social barriers. Inclusive health services remove physical, communication, financial and attitudinal barriers.

• Ageing is a normal life process, but older adults may experience increased risks of chronic disease, falls, sensory impairment and social isolation.

• Healthy ageing is supported by physical activity, nutrition, social connection, preventive care and management of chronic conditions.

• Accessibility and dignity should guide healthcare for people of all abilities and ages.

24. Health Systems, Primary Health Care and Health Promotion in Kenya

A health system includes people, institutions, financing, medicines, information, governance and service delivery. Primary health care emphasizes accessible, people-centred services, prevention, community participation and appropriate referral.

• Levels of care should work together through referral and counter-referral.

• Health promotion addresses determinants and enables healthier choices.

• Kenyan health challenges require collaboration among national and county governments, communities, health workers, schools, civil society and other sectors.

25. Health Education and Disease Prevention

Health education provides knowledge and skills that support healthier decisions. Prevention is commonly described as primary, secondary and tertiary prevention.

• Primary prevention aims to prevent disease before it occurs, e.g., vaccination and risk-factor reduction.

• Secondary prevention focuses on early detection and prompt management, e.g., appropriate screening.

• Tertiary prevention reduces disability and complications after disease is established, e.g., rehabilitation.

• Effective health communication should be accurate, understandable, culturally appropriate and respectful.

26. Contemporary Health Challenges and Future Directions

Modern health challenges are increasingly interconnected. Urbanization, population growth, climate change, antimicrobial resistance, digital technologies, misinformation and changing lifestyles can create new risks while also providing opportunities for better health.

• Antimicrobial resistance threatens effective treatment of infections and requires responsible antimicrobial use.

• Digital health can improve access to information and services but raises concerns about privacy, misinformation and unequal digital access.

• Preparedness, strong primary healthcare, surveillance, health literacy and evidence-based policy are central to resilient health systems.

REVISION QUESTIONS AND MODEL ANSWERS

Define contemporary health issues and give five examples.

Model answer: Contemporary health issues are current health problems shaped by biological, social, economic, environmental and technological factors. Examples include HIV/AIDS, NCDs, mental-health problems, malnutrition, substance use and emerging infections.

Explain the major determinants of health.

Model answer: Discuss socioeconomic conditions, education, physical environment, genetics, behaviour, healthcare access, culture, social support and public policy.

Describe the chain of infection and explain how it can be broken.

Model answer: Identify agent, reservoir, portal of exit, transmission, portal of entry and susceptible host. Prevention measures can interrupt one or more links.

Discuss five methods of preventing communicable diseases.

Model answer: Examples include vaccination, hand hygiene, safe water and sanitation, respiratory hygiene, vector control, food safety, early diagnosis and appropriate treatment.

Explain risk factors for non-communicable diseases.

Model answer: Major modifiable factors include tobacco, harmful alcohol use, unhealthy diet, physical inactivity and air pollution; non-modifiable factors include age, genetics and family history.

Differentiate primary, secondary and tertiary prevention.

Model answer: Primary prevents disease, secondary detects disease early, and tertiary limits complications/disability after disease occurs.

Discuss major contemporary mental-health challenges among young people.

Model answer: Include stress, anxiety, depression, substance use, stigma, social isolation, academic pressure and barriers to care. Explain prevention, support and referral.

Explain the importance of nutrition to health.

Model answer: Nutrition supplies energy and nutrients for growth, repair, immunity and normal body functions. Poor nutrition can cause undernutrition, deficiencies, overweight and obesity.

Discuss environmental factors that affect health.

Model answer: Cover water, sanitation, air quality, waste, housing, climate, chemical exposure and occupational hazards.

Explain the role of primary health care in improving community health.

Model answer: Primary health care emphasizes accessibility, prevention, health promotion, community participation, early treatment and referral.

QUICK REVISION: KEY TERMS

Term	Meaning
Health promotion	Actions that enable individuals and communities to improve control over health.
Prevention	Measures taken to prevent disease, detect it early, or reduce its complications.
Risk factor	A characteristic, exposure or behaviour associated with increased likelihood of a health outcome.
Immunity	The body's ability to recognize and defend against harmful organisms or substances.
Malnutrition	Deficiency, excess or imbalance of energy or nutrients.
Epidemic	Occurrence of disease cases in a community or region in excess of what is normally expected.
Pandemic	An epidemic that spreads across countries or continents and affects a large population.
Health equity	Fair opportunity for everyone to attain their full health potential, with avoidable unfair differences reduced.
Public health	Organized efforts to protect and improve the health of populations.
Primary health care	Essential, accessible, community-oriented care forming a foundation of the health system.

EXAM TIP

For essay questions, define the concept first, organize points under clear headings, explain each point briefly, give a relevant example, and finish with a concise conclusion. For short-answer questions, prioritize accurate definitions, causes/risk factors, effects, prevention and management.